

Santhosh Ravula

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SUMMARY

Software Engineer with 3+ years of experience building scalable backend systems and production-grade machine learning inference services. Strong expertise in Python-based microservices, REST API development, database optimization, and AWS cloud deployment. Experienced in integrating ML models and LLM-driven workflows into secure, high-availability backend systems. Passionate about performance optimization, system scalability, and delivering reliable AI-enabled software solutions in fintech and SaaS environments.

CORE SKILLS

Languages: Python, JavaScript, TypeScript

Backend & APIs: FastAPI, Flask, Django, RESTful APIs, Microservices Architecture, Asynchronous Processing

AI / ML Integration: TensorFlow, PyTorch, Scikit-Learn, OpenAI API, LangChain, RAG Pipelines, Prompt Engineering, Model Inference

Cloud & DevOps: AWS (EC2, Lambda, S3, SageMaker, IAM), Docker, CI/CD (GitHub Actions), Linux, CloudWatch

Databases: PostgreSQL, MySQL, MongoDB, Query Optimization, Indexing

System Design: API Scaling, Caching Strategies, Distributed Systems Concepts, High Availability, Secure API Development

EXPERIENCE

Wells Fargo

Remote – United States

Software Engineer – Machine Learning Systems

06/2025 – Present

- Designed and maintained scalable Python-based microservices supporting AI-driven fraud detection and customer insight systems
- Integrated trained TensorFlow and PyTorch models into production REST APIs, enabling real-time and batch inference workflows
- Reduced API response latency by 30% through asynchronous request handling and database query optimization
- Deployed ML inference services on AWS (EC2, Lambda, SageMaker), ensuring high availability and secure IAM-based access control
- Implemented caching and performance tuning strategies, improving system throughput during peak transaction periods
- Containerized backend services using Docker and automated deployments via CI/CD pipelines, reducing release cycle time by 25%
- Established structured logging and monitoring using CloudWatch to improve system reliability and proactive issue detection
- Collaborated with data scientists to transition experimental ML models into scalable production-ready backend services

Nexova

Hybrid – India

Software Engineer – Backend Systems

08/2021 – 07/2023

- Developed and maintained RESTful backend services using Django and PostgreSQL, supporting concurrent user workflows
- Designed normalized database schemas and implemented indexing strategies, improving query performance by 25%
- Implemented secure authentication, input validation, and API rate limiting mechanisms for production deployment
- Wrote unit and integration tests to ensure API reliability and reduce regression issues during feature releases
- Investigated and resolved production issues through debugging, log analysis, and root cause identification
- Collaborated with frontend developers and product stakeholders in Agile sprints to deliver scalable backend features

PROJECTS

Credit Card Fraud Detection Platform

Built an end-to-end ML fraud detection system using Scikit-Learn, exposing real-time model inference via FastAPI-based RESTful APIs. Deployed on AWS EC2 with Docker containerization and asynchronous processing to support concurrent transaction handling and performance optimization.

AI-Powered Task Planner (LLM Application)

Developed an LLM-driven backend application using OpenAI APIs and LangChain to convert unstructured audio into structured task workflows. Implemented prompt engineering and integrated Trello and Google Sheets APIs through secure REST endpoints for automated task orchestration.

Face Mask Detection System

Implemented a real-time computer vision solution using Python, OpenCV, and CNN models for image classification and detection. Deployed through a lightweight backend service with CI/CD-enabled updates to ensure scalable and consistent model delivery.

EDUCATION

Wright State University, Dayton, OH | Master of Science in Computer Science

TKR College of Engineering and Technology | Bachelor of Technology in Computer Science & Engineering, Hyderabad, India